INTRODUCTION TO WEB DEVELOPMENT

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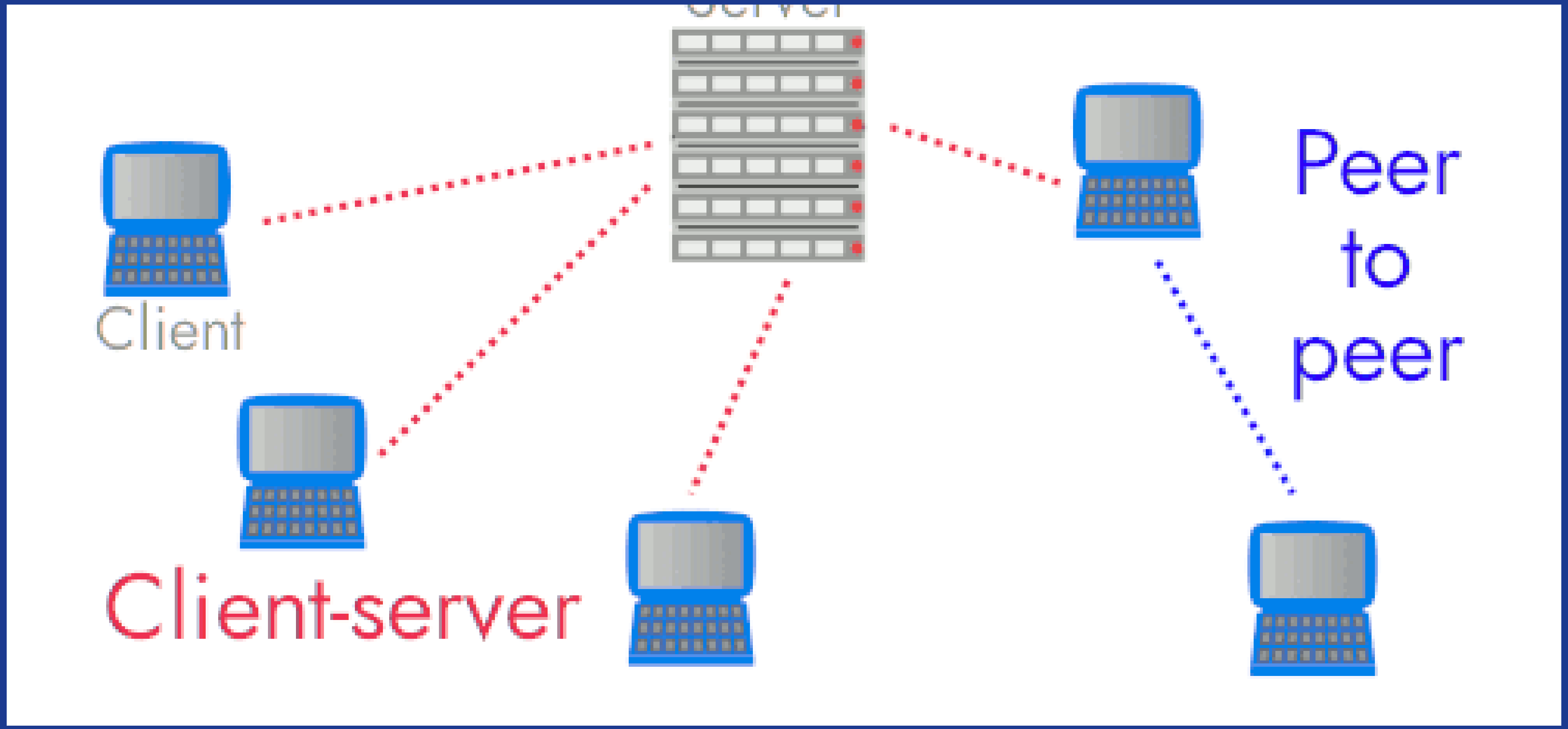
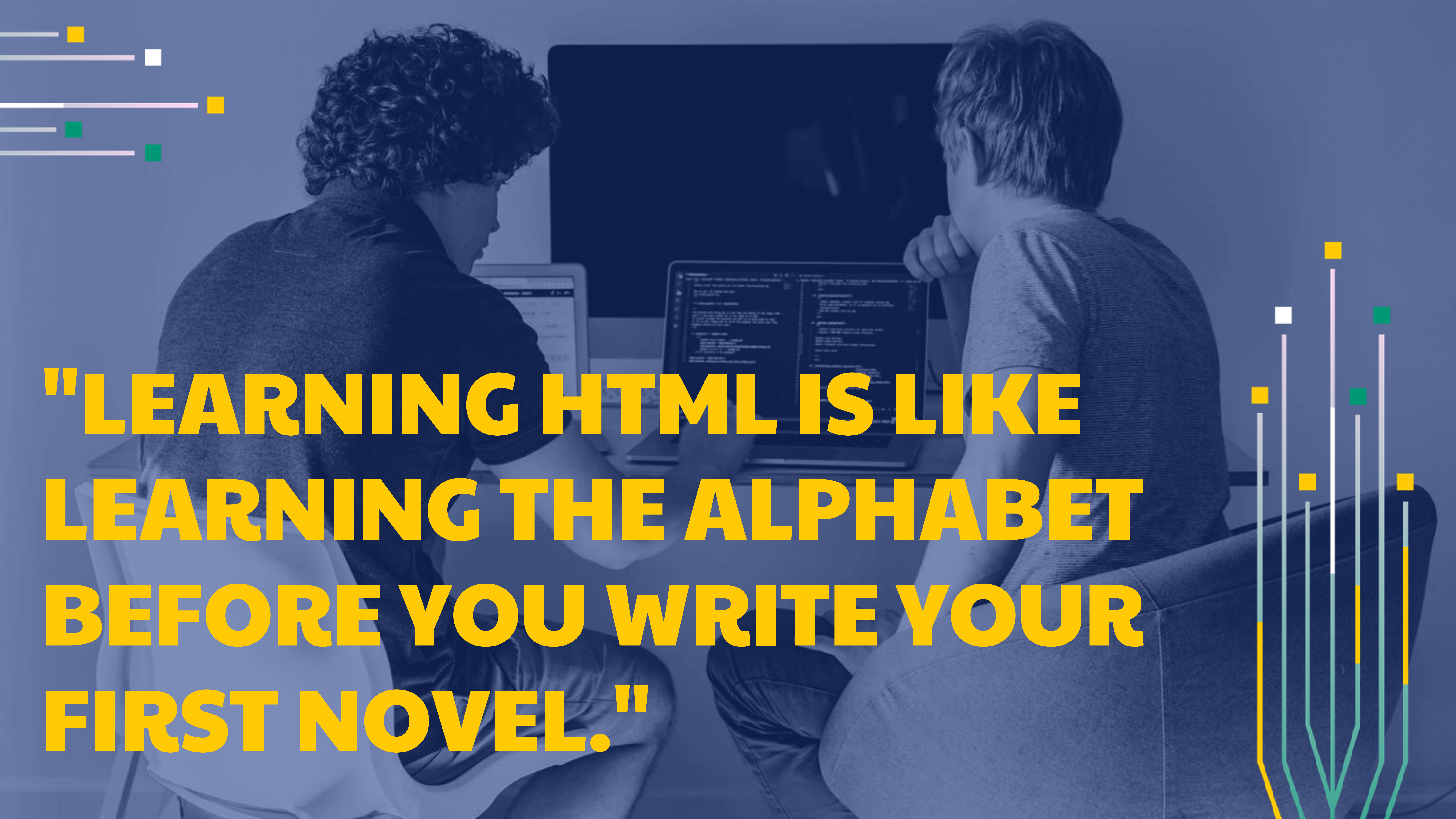
What is internet?

Interconnected network of computers around the world



What happens when you type www.google.com?

What is web development?


"LEARNING HTML IS LIKE LEARNING THE ALPHABET BEFORE YOU WRITE YOUR FIRST NOVEL."

HTML STRUCTURE

EXAMPLE

```html
<!DOCTYPE html>
<html>
 <head>
  <title>My First Web Page</title>
 </head>
 <body>
  <h1>Hello, World!</h1>
  <p>This is a simple HTML document.</p>
 </body>
</html>
```

Elements & tags

- Paragraph Tag
- Heading Tags – Wikipedia page
- Unordered lists
- Ordered lists

Anchor Tag

- Used to add links to your page
- Google

Image Tag

- Used to add images to your page
- <img src = “dog.jpeg” alt = “dog image” height=“100px” width = “100px”>

Quick Question

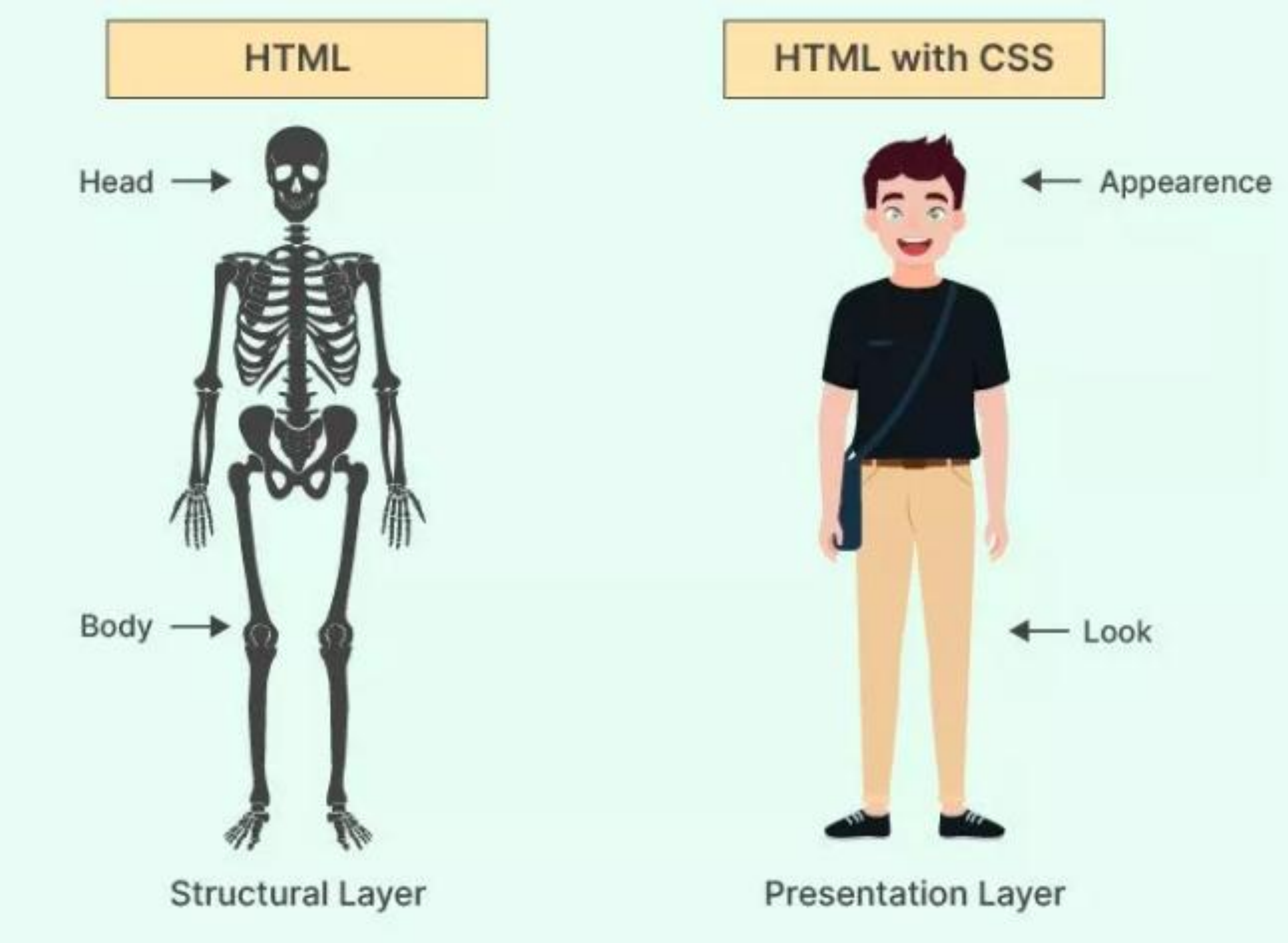
List of Fruits

- Apple see an apple
- Orange see an orange
- Mango see a mango

More Tags

- Anchor tag – inline
- Div container - ref google website
- Span

CSS


HTML
HTML with CSS
Head
Appearence
Body
Look
Structural Layer
Presentation Layer

To begin with..

- Inspect amazon website...

Inline Style

- <h1 style = “color : red”> Welcome </h1>

Style tag

```
<head>
        <style>
            h1{
                color:red;
                background-color : blue;
            }
        </style>
</head>
```

Linking HTML & CSS

- <head>

  <link rel = “stylesheet” href = “style.css”/>

  </head>

Color Property

- H1{

  color:red;

  }
- P{

  color:blue;

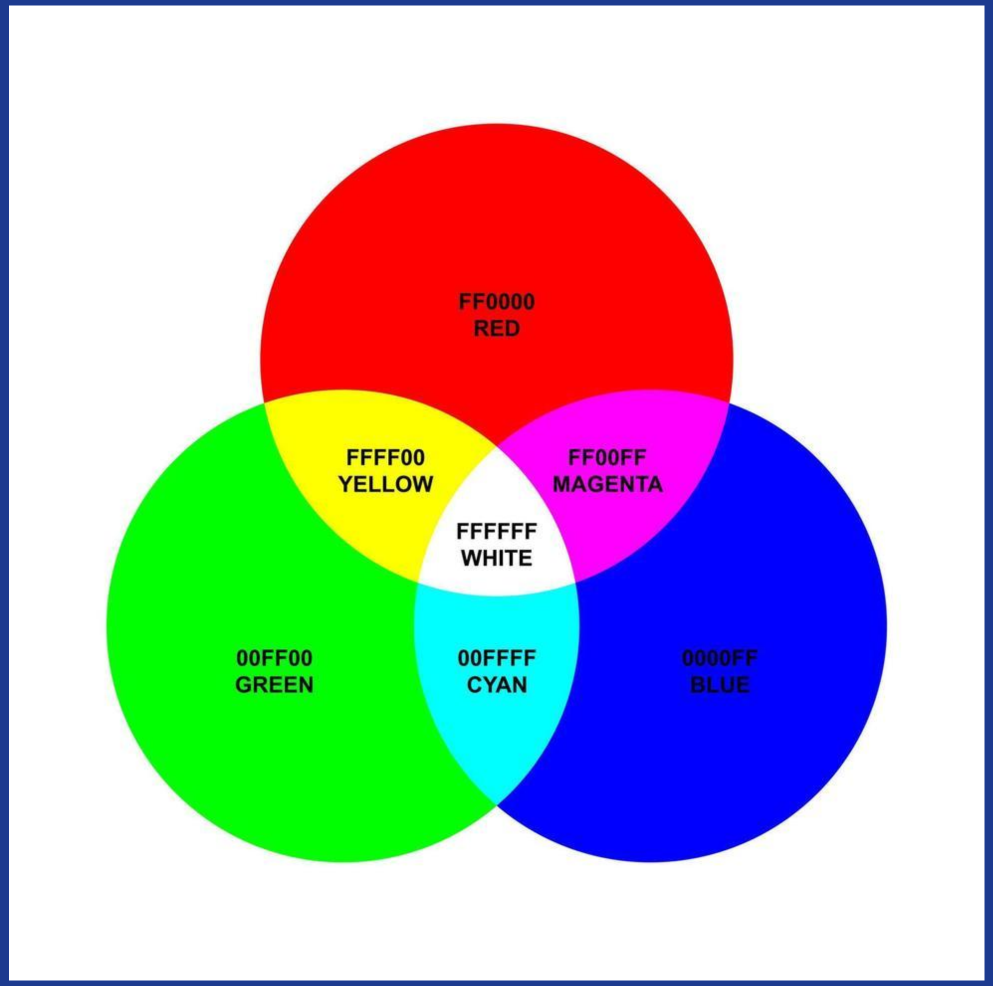
  }

BG-Color Property

- H3{
  background-color:white;
  }
- div{
  background-color:black;
  }

COLOR CODES

- Hex-codes
- Rgb



Text Properties

- Text-align
- Font-weight
- Text-decoration
- Line-height
- Letter-spacing
- Font-size
- Font family – backup fonts

Selectors

- Universal Selector
- Element selector
- Id selector
- Class selector

!important

- ```
  H1{
      background-color:blue; !important
  }
  ```
- ```
  h1,h3{
      background-color:red;
  }
  ```

Inheritance

```
<div>
    <p>
        <ul>
            <li>Apple</li>
            <li> Banana</li>
        </ul>
    </p>
</div>
```

Mini Project

Inika

Second Year, Data Science

Hi! I'm a first-year engineering student who loves coding, music, and badminton. This is my very first web page!

My Interests:

HTML
CSS
C++

Find me here: GitHub | LinkedIn

Index.html

```
<img src="photo.JPG" alt="Profile photo" width="120" height="120">

<h1 id="main-name">Inika</h1>

<h2>Second Year, Data Science</h2>

<p class="bio">

Hi! I'm a first-year engineering student who loves

coding, music, and badminton. This is my very first web page!

</p>
```

Index.html

```html
<p><strong>My Interests:</strong></p>

 <ul>

  <li class="tag">HTML</li>

  <li class="tag">CSS</li>

  <li class="tag">C++</li>

 </ul>


 <p>

  Find me here:

  <a href="https://github.com" target="_blank">GitHub</a> |

  <a href="https://linkedin.com" target="_blank">LinkedIn</a>

 </p>
```

Style.css

```css
* {

  font-family: "Poppins", Arial, sans-serif;   /* backup fonts if Poppins isn't installed */

}

/* ---- Element selector: applies to every <body> (just one, but still an element selector) ---- */

body {

  background-color: #f4f6f8;

  text-align: center;

}

/* ---- Element selector: every <h1> ---- */

h1 {

  color: #2c3e50;

  font-size: 32px;

  font-weight: bold;

  letter-spacing: 1px;

}
```

Style.css

```css
h2 {

  color: #7f8c8d;

  font-size: 16px;

  font-weight: normal;

  text-decoration: underline;

}


/* ---- Class selector: any element with class="bio" ---- */

.bio {

  color: #333333;

  font-size: 15px;

  line-height: 1.8;

  letter-spacing: 0.3px;

}
```

Style.css

```
.tag {
  color: #2c3e50;
  background-color: #ecf0f1;
  font-weight: bold;
  font-size: 14px;
  text-decoration: none;
}
/* ---- Element selector: every <a> link ---- */
a {
  color: #2980b9;
  text-decoration: none;
  font-weight: bold;
}
```

Style.css

```css
/* ---- Id selector: the one element with id="main-name" ---- */

#main-name {

  color: #e67e22;

  text-decoration: underline;

}
```

JavaScript

- Lenskart web-page
- REPL (Read-Eval-Print-Loop)
- Using the console

Variables

- A variable is simply the name of a storage location
- Example : a = 10

  age = 23

  name = John

Data Types

- Number
- Boolean
- String
- Undefined
- Null

- Using typeof
- Example : Instagram profile page

Arithemetic

- Like any other language general arithemetic operations can be performed on JavaScript.

Identifier Rules

- Can contain letters, digits, underscores and dollar signs
- Names must begin with a letter
- Names can also begin with $ and _.
- Names are case sensitive (a and A are different)
- Reserved words(keywords) cannot be used.

Boolean

- Let age = 13;
- Let isAdult = false;

Strings

- Let sentence = “this is apple”;
- Let sentence = “this is ‘apple’”;

- String indexing – 0 based
- sentence[0]
- sentence.length
- “this” + “ “+ “is” (concatenation)

Null & Undefined

- Null is the intentional absence of any object value
- A variable that has not been assigned a value is of type undefined

console.log()

- To write a message on the console

- Find the difference
- “Hello!”;
- Console.log(“Hello!”);

console.log()

- Let num = 123;

- Console.log(“num”);
- Console.log(num);
- Console.log(1+2);
- Console.log(“Hello”, “World”, num, (1+5));

Linking JS file

- <body>

  <h1> Hello </h1>

  <script src=“app.js”></script>

  </body>

Template Literals

- Let pencil = 5;
- Let eraser = 2;
- Console.log(“The total cost is:”, pencil + eraser, “rupees”);
- Let output = `The total cost is ${pencil+eraser} rupees`;
- Console.log(output);

Operators

- Arithmetic (+,-,*,/,%,**)
- Unary (++, --)
- Assignment (=, +=, -=, *=, /=)
- Comparison
- Logical

- Comparison for non-numbers based on ascii

Conditional Statements

- If-else
- Nested – if-else
- Switch case

If Statements

- If(condition){

```
        console.log("something");
  }
  console.log("After if");
```

Else - If Statements

- If(condition){

```
        console.log(“something”);
  }
  else if(condition){
            console.log(“anything”);
  }
  else if(condition){
            console.log(“statement”);
  }
```

Else Statements

- If(condition){

```
        console.log("something");
  }
  else( condition){
        console.log("anything");
  }
```

Nested - if Statements

- Let marks = 45;
- If(marks>=33){

```
        console.log("Pass");
        if(marks >= 80) {
              console.log("Grade : O");
        } else  {
              console.log("Grade: A");
        }
  } else {
        console.log("Better Luck next time!");
  }
```

Truthy & Falsy

- If(true){

  console.log("It has true value");

  }

  else{

  console.log("It has false value");

  }

- Falsy values – false, 0,-0, "", null , undefined, NaN

Switch - Case

- Let color = “red”;
- Switch(color) {

    case “red”: console.log(“stop”);

        break;

    case “yellow” : console.log(“slow down”);

        break;

    case “green” : console.log(“GO”);

        break;

    default : console.log(“Broken Light”);

  }

Alert & Prompt

- Alert displays an alert message on the console
- Alert(“something is wrong!”);

- Prompt displays a dialogue box that asks user for input
- Prompt(“please enter your roll no.”);

What is git?

Free and Open source version control system

- Helps to track history
- Helps to collaborate

What is github?

Websites where we host repositories online

Using GitHub

- Lets create our first repository
- Write and delete content
- Check commit history

Using Git

- Command line
- IDE/Code Editors (like VS Code)
- Graphical User Interfaces (like GitKraken)

Using Git

- Git –version
- Git

Configuring Git

- Git config –global user.name “NAME”
- Git config –global user.email “EMAIL”
- Git config --list

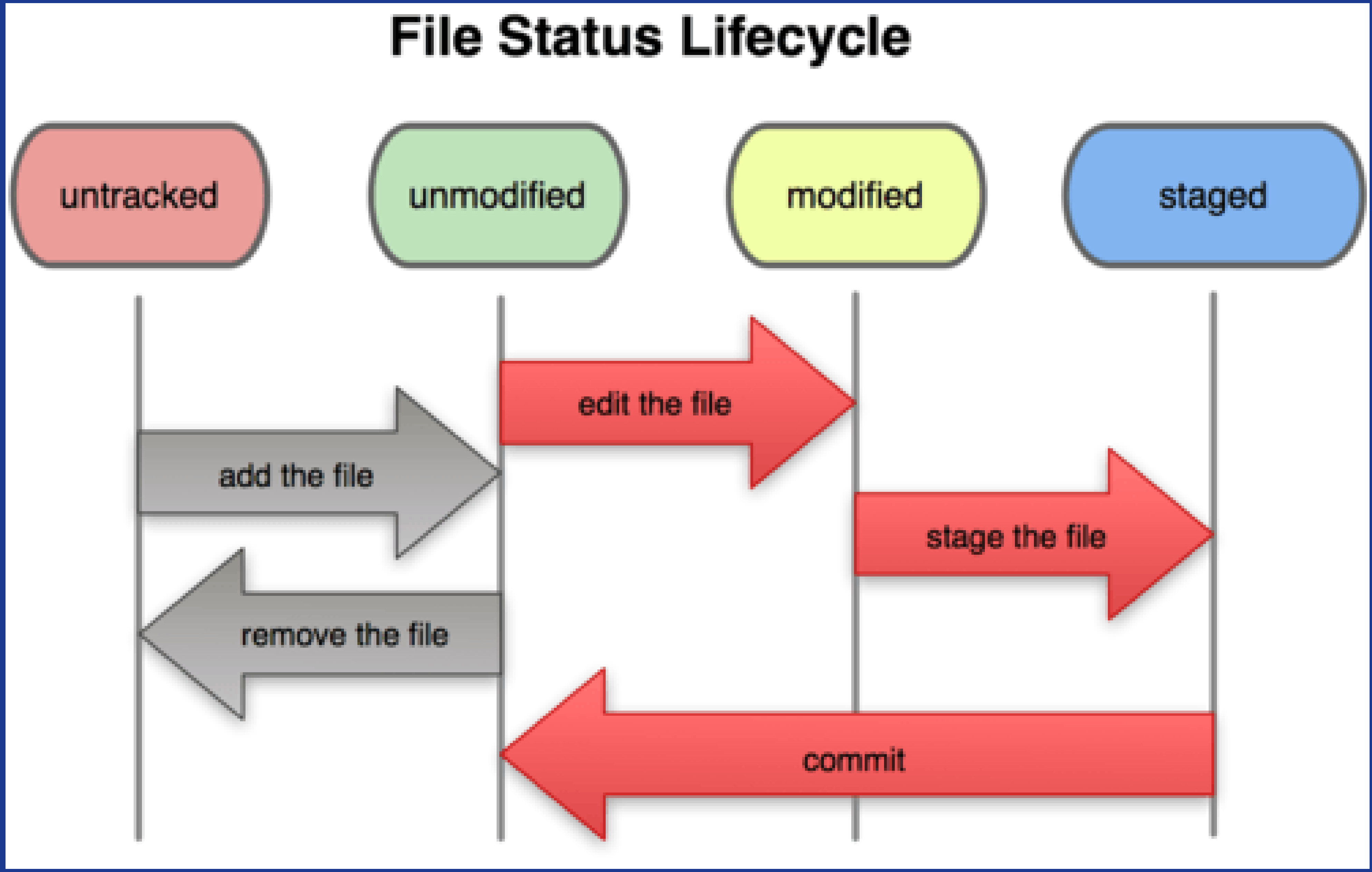
Basic Commands

- Clone – cloning a repository on our local machine
- Git clone <-some link->

Basic Commands

- Status – displays the state of the code
- Git status

File Status LifeCycle



Basic Commands

- add – adds new or changed files in working directory to staging area
- Git add <-file name->

Basic Commands

- commit – it is th record of change
- Git commit –m “some message”

Basic Commands

- push – upload local repo content to remote repo
- Git push origin main

Init command

- Init – used to create a new git repo
- Git init

- Git remote add origin <-link->
- Git remote –v (to verify remote)
- Git branch (to check branch)
- Git branch –M main (to rename branch)
- Git push origin main

Sources

https://github.com/Inika15/Skill-lab-day1-webdev-resources

GitHub repository

Lets Connect!

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THANK YOU